

Credible on Paper? Green Banking Policy, Access to Finance, and Institutional Capacity in the Greening of Firms Worldwide

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Abstract

Green banking policy frameworks coordinated through the Sustainable Banking and Finance Network (SBFN) have been adopted across more than sixty emerging and developing economies since 2012, on the premise that supervisory guidelines directing banks to price environmental risk into lending will strengthen the link between firms' access to credit and their green investment behaviour. We test this premise using two independent World Bank Enterprise Survey samples. The primary sample harmonizes the Bank's full 2018–2020 Green Economy Module rollout across 41 economies (28,042 firms) on a rich seven-item green-practice-adoption outcome, matched to each economy's SBFN policy-adoption status and Worldwide Governance Indicators regulatory quality as of its survey year. Because SBFN status is fixed within country in a single-wave cross-section, a classical country-fixed-effects specification cannot identify it or its interactions; we show this explicitly, then re-estimate the relationship with a Bayesian hierarchical model (country-varying slopes) and a causal forest that impose no linear-interaction functional form. All three approaches agree: bank credit access is robustly associated with green practice adoption (average marginal effect of 13 percentage points in

the baseline specification), but neither SBFN policy adoption nor regulatory quality moderates it—both institutional variables instead shift the general level of green adoption directly. We then subject this finding to the sharpest test we can construct: a second, independently assembled sample of 162 further economies (89,797–100,115 firms depending on specification), spanning 2021–2026 and reaching, for the first time in this literature, into high-income economies such as Australia, Germany, and Japan, built from a World Bank standardized cross-country database not previously exploited for this purpose. On a narrower CO2-emissions-monitoring outcome and with country fixed effects the primary sample’s single-wave design cannot support, credit access remains significant while both SBFN and regulatory-quality interactions remain statistically indistinguishable from zero—the identical qualitative pattern, on a different outcome, a different set of economies, and a materially sharper identification strategy. The causal forest’s feature-importance decomposition attributes the bulk of genuine effect heterogeneity in the primary sample (75.6%) to firm size rather than to either institutional moderator (regulatory quality 19.9%; SBFN status 2.6%). One finance-instrument-specific check finds a nominally significant negative credit $\times$ SBFN interaction when finance access is measured via overdraft rather than term credit, though this does not survive a Benjamini–Hochberg correction for multiple comparisons across the full family of interaction tests reported in the paper, and we treat it as suggestive rather than established. We interpret the combined evidence as more consistent with green banking policy’s binding constraint operating through firm-level absorptive capacity than through the country-level regulatory capacity its own institutional design targets, though

we report this as our best current account of an auxiliary, exploratory finding rather than as a settled result.

Keywords: green banking policy, access to finance, institutional complementarity, firm-level green investment, causal forest, hierarchical Bayesian model

JEL: G21, O16, Q56, D22, C11

1. Introduction

Since 2012, more than sixty central banks, banking regulators, and financial-sector regulatory bodies across the emerging and transition world have signed on to a coordinated effort to green their financial systems, adopting national guidelines that direct commercial banks to price environmental and climate risk into lending decisions and to channel credit toward cleaner production. On paper, this represents one of the most ambitious diffusions of environmental financial regulation outside the OECD: economies as different in income, banking-sector depth, and political regime as Bangladesh, Morocco, Mongolia, and Indonesia now report, to the International Finance Corporation’s Sustainable Banking and Finance Network (SBFN), formal sustainable-finance roadmaps, green-taxonomy pilots, or mandatory environmental-risk disclosure rules for lenders. Whether that adoption changes what actually happens on a factory floor—whether a firm that can get a bank loan is any more likely to install a wastewater treatment system, switch to a more efficient boiler, or measure its own emissions—is a separate and considerably harder question, and it is the one this paper answers.

The premise that formal adoption of a policy is not the same as its op-

eration in practice is, of course, not new to institutional economics. A large literature on state capacity and policy credibility has long argued that the same law, regulation, or guideline can produce entirely different real-economy outcomes depending on the administrative and judicial infrastructure available to enforce it (La Porta et al., 1998; Acemoglu et al., 2001). Applied to green finance specifically, a parallel body of work has documented that green credit guidelines in China measurably reshape lending allocation and firm environmental performance (Li et al., 2023; Huang et al., 2023; Dai et al., 2025), that firms’ green investment responds to the financing constraints they face (Ghosh and Dutta, 2022; Costa et al., 2024; De Haas et al., 2025), and that the World Bank Enterprise Surveys’ own Green Economy module is a workable lens on firm-level environmental behaviour across dozens of countries at once (Asif et al., 2025; Phan, 2026). What is missing is the connection between these two literatures: a direct test, at firm level and across a wide cross-section of the very economies the green-banking-policy movement targets, of whether the credit-to-green-investment channel that these guidelines are designed to activate is actually stronger where the guidelines exist—and whether that depends on the state capacity to make the guideline credible in the first place.

This paper closes that gap using firm-level data from two independently assembled samples of World Bank Enterprise Survey (WBES) waves. The primary sample covers 47 country-years fielded between 2018 and 2023 that carry the Bank’s full Green Economy module—a battery of questions on emissions monitoring, energy audits, and the adoption of climate-friendly production measures—alongside the Surveys’ standard access-to-finance module; a

44 second, global sample extends this to 162 further country-years fielded be-
 45 tween 2021 and 2026, reaching for the first time in this literature into high-
 46 income economies with no SBFN involvement at all (Section 4). We match
 47 each economy to its Sustainable Banking and Finance Network policy status
 48 as of its survey year, distinguishing genuine ex-ante policy adopters from
 49 economies that joined only afterward, and to the World Bank’s Worldwide
 50 Governance Indicators as a measure of the regulatory state capacity avail-
 51 able to make that policy credible. Econometrically, we depart deliberately
 52 from the pooled fixed-effects specification that has become the default in
 53 this literature. Because SBFN status is fixed within country in a single-wave
 54 cross-section, a country fixed effect absorbs it entirely, and a naive interaction
 55 term estimated by pooled logit—which we report first, as a baseline, precisely
 56 to make this limitation visible rather than to hide it—returns no detectable
 57 policy-moderation effect at all. We show that this null result is not evidence
 58 of an absent relationship so much as a symptom of forcing a fundamentally
 59 hierarchical question (does a level-2, country-level policy variable reshape
 60 a level-1, firm-level financing relationship?) into a flat, single-level model
 61 with too few country clusters to lean on for inference (Cameron and Miller,
 62 2015). We then re-estimate the same relationship with tools built for exactly
 63 this structure: a Bayesian hierarchical logistic model with country-varying
 64 slopes (Gelman and Hill, 2007; Gelman et al., 2020), a causal-forest / double-
 65 machine-learning (DML) estimator (Wager and Athey, 2018; Chernozhukov
 66 et al., 2018; Athey et al., 2019; Athey and Imbens, 2019) that lets the treat-
 67 ment effect of credit access vary flexibly across the institutional landscape
 68 rather than through a single estimated coefficient, and, on the much larger

69 global sample, a country-fixed-effects specification that the primary sample's
70 single wave cannot support.

71 This paper contributes to the literature in five respects. First, it pro-
72 vides what is, to our knowledge, the first direct empirical test of whether
73 SBFN-style green banking policy adoption changes the firm-level relation-
74 ship between finance and green investment, rather than only the aggregate
75 volume of green lending reported by banks themselves—a firm-level, demand-
76 side complement to the bank-level, supply-side evidence that dominates the
77 existing green-credit-policy literature. Second, it extends the World Bank
78 Enterprise Survey's Green Economy module—used until now mostly within
79 single countries or single regional waves—into a harmonized 41-country pri-
80 mary sample spanning Europe, Central Asia, the Caucasus, and the Middle
81 East and North Africa, giving the paper a breadth of institutional varia-
82 tion that a single-country or single-region design cannot offer. Third, and
83 unusually for this literature, we subject the primary sample's central find-
84 ing to an independently assembled second test: a 162-economy global sam-
85 ple built from a World Bank standardized cross-country database spanning
86 2021–2026 that, to our knowledge, no published study of green banking pol-
87 icy has yet exploited, extending coverage for the first time into high-income
88 economies entirely outside the SBFN's own remit and allowing a country-
89 fixed-effects specification the primary sample's single-wave design cannot
90 support. That the same qualitative result—a robust finance-green channel,
91 no detectable institutional moderation—survives this independent replica-
92 tion is, we think, considerably more informative than either sample could
93 be alone. Fourth, methodologically, the paper demonstrates—with the same

94 underlying question tested twice, on two different samples, under four differ-
 95 ent estimators—exactly where and why a classical fixed-effects approach fails
 96 to detect (or, when country fixed effects are usable, correctly fails to find) an
 97 institutionally-conditioned effect, offering applied researchers working with
 98 similarly structured cross-country firm data a concrete illustration of when
 99 the extra modelling investment is worth making and when a null result is
 100 simply a null result. Fifth, the paper redirects the question of which system-
 101 level margin governs green-credit-policy effectiveness: having ruled out a
 102 country-institutional interaction across four estimators and two samples, we
 103 show (Section 3.4) that the effect heterogeneity that does exist is concen-
 104 trated at the level of firm size rather than country institutions, relocating
 105 the operative “system condition” in the Policy Design $\times$ System Conditions
 106 $\rightarrow$ System Outcomes logic this special issue organizes itself around from the
 107 country-regulatory level to the firm-scale level—a mechanism we introduce
 108 as an auxiliary, exploratory hypothesis in Section 3.4 and demonstrate em-
 109 pirically in Section 5.4 rather than treat as a pre-registered finding, and are
 110 correspondingly explicit about the identification limits (Section 4) that a
 111 purely cross-sectional design, however extensively replicated, still carries.

112 The paper also speaks directly to a growing body of recent scholarship in
 113 this journal on the institutional and financial determinants of firm and bank
 114 behaviour in emerging and transition economies. Yan et al. (2024) study
 115 how environmental protection taxes reshape green productivity among Chi-
 116 nese listed firms; Barra and Falcone (2026) examine how institutional factors
 117 condition environmental performance across a global sample of economies;
 118 Heyert and Weill (2025) and Bach et al. (2025) use firm- and bank-level data

119 across dozens of countries to study, respectively, trust-driven financial inclu-
120 sion and credit constraints’ effect on corporate tax behaviour in Vietnamese
121 firms; Li et al. (2026) study how shadow-banking engagement reshapes trade-
122 credit financing among Chinese non-financial firms; and Horvath et al. (2025)
123 apply Bayesian model averaging to the cross-country determinants of finan-
124 cial development, a methodological precedent we build on directly in adopting
125 a Bayesian estimation strategy in Section 4. This paper sits at the intersec-
126 tion of these threads—institutions, green performance, and firm-level financ-
127 ing behaviour, studied jointly rather than pairwise—and, to our knowledge,
128 is the first among them to test whether a specific, dateable policy-adoption
129 event (SBFN membership) moderates the finance-green relationship at firm
130 level across a comparably broad cross-section of adopting economies.

131 The remainder of the paper is organized as follows. Section 2 sets out
132 the institutional background of the SBFN framework and the concepts and
133 trends in green banking policy adoption across our sample. Section 3 devel-
134 ops the paper’s three hypotheses from the state-capacity and institutional-
135 complementarity literature. Section 4 describes the primary and global sam-
136 ples, variable construction, and the empirical strategy for each. Section 5
137 presents baseline, hierarchical, and heterogeneous-effect results on the pri-
138 mary sample, followed by the independent global-sample replication. Sec-
139 tion 6 discusses the findings’ economic magnitude and their implications for
140 how green credit policy is designed and sequenced, and Section 7 concludes.

141 **2. Institutional background: green banking policy in emerging and** 142 **transition economies**

143 The Sustainable Banking and Finance Network (SBFN)—established in
144 2012 as the Sustainable Banking Network and renamed in 2019 to reflect an
145 expanded mandate covering both banking and non-bank financial regulation—
146 is the principal vehicle through which financial-sector regulators and banking
147 associations in emerging and developing economies have coordinated the de-
148 sign of national sustainable-finance policy. Hosted and technically supported
149 by the International Finance Corporation, the network now counts roughly
150 seventy member institutions (central banks, banking regulators, and banking
151 or securities associations) drawn overwhelmingly from Sub-Saharan Africa,
152 East Asia and the Pacific, Europe and Central Asia, Latin America and
153 the Caribbean, the Middle East and North Africa, and South Asia (Inter-
154 national Finance Corporation / Sustainable Banking and Finance Network,
155 2025). Notably for our sample, the network’s membership is essentially dis-
156 joint from the European Union: not one of the sixteen EU member states in
157 our WBES sample appears on any SBFN membership roster we could locate,
158 consistent with the network’s explicit mandate to serve markets outside the
159 reach of the EU’s own sustainable-finance architecture (the Sustainable Fi-
160 nance Disclosure Regulation and the EU Taxonomy, both of which post-date
161 most of our survey waves in any case).

162 Membership itself, however, is only the network’s coarsest measure of
163 policy adoption. SBFN’s own Multi-Tiered Progress Framework—the in-
164 strument used in its periodic Global Progress Reports (2018, 2019, 2021,
165 2023–2024) and country-level Progress Report addenda—scores each mem-

166 ber jurisdiction’s actual regulatory output against several policy pillars, most
 167 centrally environmental-and-social risk management and, in later report vin-
 168 tages, ESG disclosure integration and green-finance-flow measurement, and
 169 places each jurisdiction on a progression from an initial “Commitment” or
 170 “Formulating” stage, through “Developing” or “Implementation,” to “Advanc-
 171 ing” and, for the most mature frameworks, “Consolidating.” The variation
 172 this produces across our sample is considerable and is itself part of the paper’s
 173 identifying variation, not just a robustness afterthought. Mongolia’s Bank
 174 of Mongolia was a founding SBFN member in 2012 and had already reached
 175 the network’s “Advancing” implementation stage by the time of the country’s
 176 2019 Enterprise Survey; Morocco’s Bank Al-Maghrib, a member since 2014,
 177 sat at the same stage two years later. Bangladesh Bank, also a founding 2012
 178 member, had progressed by its 2022 Enterprise Survey wave into the net-
 179 work’s ESG-integration pillar at an “Advancing” classification, while India’s
 180 Reserve Bank/Indian Banks’ Association, a later (2016) entrant, remained
 181 at a “Developing” commitment-pillar stage at its own 2022 survey. Jordan
 182 and the Kyrgyz Republic, both members since 2016 and 2018 respectively,
 183 had progressed only as far as an initial “Commitment” stage by their 2019
 184 survey waves. A further, and for identification purposes equally important,
 185 group of economies in our sample joined SBFN only after their Enterprise
 186 Survey was fielded—Kazakhstan (2021), Serbia and Ukraine (2020), Tajik-
 187 istan (2023), and most of the Western Balkan and South Caucasus economies
 188 in our sample (2021–2024)—and are accordingly coded as non-adopters as of
 189 their survey year, even though several are now members. This ex-ante cod-
 190 ing is deliberate: it is the policy environment a firm actually operated under

191 at the moment its Green Economy module responses were recorded that is
192 relevant to our test, not the policy environment that jurisdiction eventually
193 adopted.

194 The economic logic behind the SBFN framework, and the reason it is a
195 plausible channel for firm-level green investment rather than a purely sym-
196 bolic commitment, rests on a straightforward supervisory mechanism: once
197 a banking regulator formally adopts environmental-and-social risk manage-
198 ment guidelines, commercial banks under its supervision face a compliance
199 incentive to price environmental risk into individual lending decisions and,
200 in many of the more advanced frameworks, to report a green-lending share
201 to the supervisor directly. Existing evidence on China’s much-studied green
202 credit guidelines documents exactly this channel operating at bank and firm
203 level—reduced lending to high-polluting borrowers and improved environ-
204 mental performance among firms that retain access to credit (Li et al., 2023;
205 Huang et al., 2023; Dai et al., 2025; Gao et al., 2022)—and a smaller liter-
206 ature on Bangladesh’s own long-running green-banking guidelines points in
207 the same direction (Khairunnessa et al., 2021). What neither literature has
208 established is whether this channel is detectable across a broad, heteroge-
209 neous cross-section of SBFN-style adopters at the firm level using a common
210 survey instrument, nor whether the strength of the channel depends on the
211 regulatory capacity available to enforce the underlying guideline—the two
212 questions this paper turns to next.

213 **3. Literature review and hypothesis development**

214 *3.1. Access to finance and firm-level green investment*

215 A firm’s decision to install pollution-abatement equipment, retrofit ma-
216 chinery for energy efficiency, or absorb the upfront cost of an emissions-
217 monitoring system is, first and foremost, an investment decision, and in-
218 vestment decisions are shaped by whether a firm can finance them. The
219 theoretical case is close to a standard credit-constraint story: green invest-
220 ments of the kind captured in the WBES Green Economy module (heat-
221 ing and cooling upgrades, on-site renewable generation, waste-minimization
222 retrofits, energy-management systems) typically carry a large upfront capi-
223 tal outlay against a payback stream realized only over several years, which is
224 exactly the profile of investment most sensitive to a firm’s access to external
225 finance rather than to retained earnings alone. Firm-level evidence bears
226 this out directly. Ghosh and Dutta (2022), using a panel of Indian manu-
227 facturing firms, show that credit-constrained firms are measurably less likely
228 to adopt cleaner production practices even when they report an equivalent
229 willingness to comply with environmental regulation. De Haas et al. (2025)
230 document, across a large cross-country firm sample, that managerial and fi-
231 nancial barriers operate as distinct but jointly binding constraints on firms’
232 green transition investments, and Kalantzis et al. (2022) reach a closely re-
233 lated conclusion using EBRD transition-economy survey data, finding that
234 firms’ green investment is driven by a mix of financing conditions and cli-
235 mate exposure rather than by climate motivation alone—precisely the joint
236 finance-and-institutions framing this paper tests directly. Costa et al. (2024)
237 and Truong (2026) find, respectively in OECD and Vietnamese data, that

238 financing constraints blunt firms’ green investment response to environmen-
 239 tal policy pressure that would otherwise induce it, while Aristei and Gallo
 240 (2023) show that firms with better access to credit are both more likely to
 241 have adopted green management practices before a shock and better able
 242 to sustain them through one. Ullah (2025) narrows this further to unlisted
 243 SMEs specifically, showing that financial constraints bind corporate environ-
 244 mental responsibility at smaller firm scales—a firm-size channel this paper’s
 245 own causal-forest evidence (Section 5) also implicates, though, as we report
 246 there, without recovering a clean monotonic size gradient of its own. Siewers
 247 et al. (2024) extend the firm-level environmental-performance literature to
 248 a global-value-chain setting, showing that a firm’s position in international
 249 production networks shapes its environmental performance net of domes-
 250 tic financing conditions, and Bambe et al. (2026) show that environmental
 251 policy stringency itself reshapes firm efficiency in developing economies, un-
 252 derscoring that the firm-level margin, not only the aggregate or bank-level
 253 one, is where much of this literature’s true test now lies. Taken together, this
 254 literature converges on a simple first-order proposition, which any test of a
 255 green-credit-policy channel must first establish before asking whether policy
 256 moderates it:

257 **H1.** Firms with access to formal bank credit (a credit line or
 258 loan from a financial institution) are more likely to have adopted
 259 green production practices than otherwise-similar firms without
 260 such access.

261 *3.2. Does green banking policy strengthen the finance-to-green-investment*
262 *channel?*

263 H1 describes a channel, not a policy. The entire premise of the SBFN
264 framework and its national analogues—and of the much larger literature on
265 green credit guidelines in China specifically (Li et al., 2023, 2024; Huang
266 et al., 2023)—is that a supervisory mandate directing banks to price envi-
267 ronmental risk into lending, or to preferentially finance green activity, should
268 widen the gap in green-investment behaviour between firms with and with-
269 out bank access relative to a counterfactual world without the policy: banks
270 in adopting jurisdictions have a compliance incentive to seek out, and price
271 favourably, exactly the borrowers undertaking the kind of investment our
272 outcome variable measures, while credit-constrained firms in the same ju-
273 risdiction see no comparable improvement in their financing terms. This
274 is a moderation hypothesis, not a simple main-effect one, and it is worth
275 being explicit that most of the existing supporting evidence—again, over-
276 whelmingly from China (Dai et al., 2025; Gao et al., 2022) plus a smaller
277 literature on Bangladesh (Khairunnessa et al., 2021), Vietnam (Nguyen and
278 Phan, 2025), and cross-country fintech-credit channels (Tran et al., 2026)—
279 is drawn from single-country studies of a single, unusually well-resourced
280 green-credit regime. Whether the same interaction holds across the far more
281 heterogeneous set of jurisdictions that have adopted an SBFN-style frame-
282 work, many of them at a far earlier and less resourced stage of implementation
283 than China’s, is untested. This motivates:

284 **H2.** The positive association between bank credit access and
285 green-practice adoption (H1) is stronger in economies that had,

286 as of the survey year, adopted an SBFN sustainable-finance policy
287 framework than in economies that had not.

288 *3.3. Institutional complementarity: does policy credibility depend on state*
289 *capacity?*

290 A green banking guideline is only as effective as the supervisory apparatus
291 behind it. This is the institutional-complementarity argument at the centre of
292 the comparative-institutions, law-and-finance, and state-capacity literatures.
293 Hall and Soskice (2001)’s varieties-of-capitalism framework is the founda-
294 tional statement of the underlying logic in its most general form: formally
295 similar policy instruments produce systematically different outcomes depend-
296 ing on how they interlock with the surrounding institutional architecture—
297 a regulatory mandate that functions as intended within one configuration
298 of complementary institutions may function quite differently, or not at all,
299 transplanted into another. Applied more narrowly to financial regulation
300 specifically, the same logic runs through the law-and-finance and state-capacity
301 literatures: La Porta et al. (1998) and the subsequent finance-and-growth
302 literature (Levine, 2005; Demirgüç-Kunt and Maksimovic, 1998; Beck et al.,
303 2005, 2008) established that formally identical financial-contracting rules pro-
304 duce very different real-economy financing outcomes depending on the quality
305 of the legal and regulatory institutions that enforce them, while Acemoglu
306 et al. (2001) showed the same logic operating at the level of a country’s en-
307 tire institutional endowment. Applied to our setting, the argument is direct:
308 a bank supervisor’s environmental-risk guideline can only reshape lending
309 behaviour to the extent that the supervisor has the regulatory capacity—
310 examination authority, credible sanctions, reliable disclosure enforcement—

311 to make compliance with it more than nominal. Where that capacity is
312 weak, an SBFN framework may exist on paper (a formal “Commitment” or
313 “Formulating” stage classification, in the network’s own terminology) with-
314 out meaningfully altering bank lending behaviour on the ground, precisely
315 the state-capacity/policy-credibility distinction that motivates this special
316 issue. This is also the logic behind the small but growing strand of work on
317 institutions and environmental performance published in this journal specifi-
318 cally: Barra and Falcone (2026), studying a global sample of economies, find
319 that institutional-quality indicators condition environmental performance in
320 ways that are not reducible to income level alone, a result our design is built
321 to test at the firm rather than the country level. We proxy regulatory ca-
322 pacity with the World Bank’s Worldwide Governance Indicators Regulatory
323 Quality estimate, matched to each economy’s survey year, and hypothesize
324 a three-way interaction:

325 **H3.** The policy-moderation effect described in H2 is itself in-
326 creasing in regulatory quality: SBFN adoption strengthens the
327 credit-to-green-investment channel most where the supervising
328 jurisdiction has the institutional capacity to make the underly-
329 ing guideline credible, and least—potentially not at all—where it
330 does not.

331 Hypotheses H2 and H3 are, by construction, statements about how a
332 country-level variable reshapes a firm-level relationship—precisely the two-
333 level structure that Section 4 argues a classical single-level fixed-effects re-
334 gression is poorly suited to recover with a country-cluster count in the low

forties, and that motivates the hierarchical and machine-learning estimators we turn to after reporting that classical benchmark.

3.4. An auxiliary hypothesis: firm-level absorptive capacity as an alternative margin

H2 and H3 are both, in the end, statements that a *country-level* characteristic conditions the strength of the credit-to-green channel. A separate strand of the financing-constraints literature suggests the conditioning variable that matters most may instead sit at the *firm* level: Ullah (2025) shows that financial constraints bind corporate environmental responsibility specifically at smaller firm scales, consistent with a standard fixed-cost logic in which the up-front capital outlay a green retrofit requires (Section 3) is proportionately heavier, and therefore harder to absorb even once financed, for a smaller firm than a larger one. We did not set out to test this as a fourth ex-ante hypothesis on a par with H1–H3; it emerged from the causal forest’s feature-importance decomposition (Section 5.4), which we report honestly here as an auxiliary, exploratory hypothesis motivated by that same fixed-cost logic rather than one we pre-specified:

H4 (auxiliary, exploratory). Whatever heterogeneity exists in the strength of the credit-to-green-adoption channel is more attributable to firm size than to either country-level institutional moderator (SBFN status or regulatory quality).

We flag its exploratory status explicitly because it changes how its evidence should be read: Section 5.4’s support for H4 is genuine and, we think,

informative, but it is evidence discovered during analysis rather than confirmed against a pre-registered prediction, and the same section reports honestly that it does not yet extend to a clean, monotonic size gradient. H4 also connects this paper’s evidence to the just-transition and SME-financing literature’s broader concern with who is left behind when a policy instrument operates on an extensive banking margin rather than on the firm-scale margin that determines whether a financed firm can absorb a green investment’s fixed costs at all—a distributional dimension of green-credit-policy design this paper’s data can only gesture toward, not test directly, but which we think is a natural extension for future work using firm-level data with a continuous size measure and a larger, more granular firm-size distribution than the Enterprise Surveys’ own three-category strata provide.

4. Data and methodology

4.1. Data sources and sample construction

The paper’s firm-level data come from the World Bank Enterprise Surveys (WBES), a standardized, stratified-random-sample survey of formal-sector, non-agricultural, non-extractive private firms with at least five employees, conducted by the World Bank’s Enterprise Analysis Unit. Our primary analysis sample is the World Bank’s own harmonized cross-country file combining the 41 Europe and Central Asia (ECA) and Middle East and North Africa (MENA) economies surveyed under a single, identical questionnaire between 2018 and 2020, the survey wave in which the Bank’s Green Economy module—a battery of questions on emissions monitoring, energy-consumption targets, and the adoption of specific climate-friendly production

measures—was fielded in full to every respondent. This yields 28,042 firm-level observations across 41 economies (Table 2). We supplement this primary sample with 15,556 additional firm observations from six further economies (Bangladesh 2022, India 2022, Indonesia 2023, Peru 2023, Philippines 2023, Timor-Leste 2021) in which a shortened, partially randomized version of the Green Economy module was fielded; because the exact item wording and randomized sub-module assignment differ across these six economies, we do not pool them into the main harmonized outcome variable, and instead use them as an out-of-region external-validity check on a narrower, honestly-matched pair of indicators (CO2-emissions monitoring, available in five of the six; waste-minimization adoption, available in five of the six, with different specific economies covered by each), reported separately in Section 5.

The primary sample, while broad by the standards of the existing green-credit-policy literature, is not the full universe of WBES economies carrying *any* Green Economy content, and we build a second, independent sample to exploit the rest of it. The World Bank also maintains a standardized cross-country database (2006–2026) that harmonizes a common set of core variables across every Formal Sector Enterprise Survey wave it has ever fielded, 292,314 firm-level observations across 471 country-years. Within it, 160 country-years fielded between 2021 and 2026 — reaching, notably, into high-income economies entirely outside the SBFN’s remit such as Australia, Canada, Germany, and Japan — carry a harmonized minimal Green Economy item set: climate-damage exposure and, more relevantly for our outcome construct, CO2-emissions monitoring, alongside the same standard access-to-finance module (credit line/loan, overdraft, perceived finance ob-

407 stacle) used in the primary sample. We supplement this with Bangladesh
 408 2022 and Indonesia 2023 — two economies whose Green Economy responses
 409 used an alternative randomized-module variable naming convention and so
 410 are absent from the standardized database’s harmonized columns, but which
 411 we can recover from our own extraction of the underlying country files —
 412 for a combined global sample of 162 economies and 100,115 firms (89,797
 413 with complete data on all covariates). Firm size, sector, and productivity
 414 controls for this sample are drawn from the Bank’s companion Productivity
 415 Database, merged by firm identifier. We refer to this as the *global sample*,
 416 and to CO2-emissions monitoring as its outcome variable throughout; it is
 417 the paper’s vehicle for testing whether the primary sample’s central find-
 418 ing replicates independently, at far greater scale, and under an identification
 419 strategy — country fixed effects — that the primary sample’s single-wave
 420 design cannot support (Section 4).

421 Country-level data are drawn from two sources for both samples. For the
 422 primary sample’s 47 economy-years, SBFN policy-adoption status, join year,
 423 and progression-framework stage as of each economy’s survey year are com-
 424 piled from the Sustainable Banking and Finance Network’s Global Progress
 425 Reports (2018, 2019, 2021, 2023–2024), country-level Progress Report ad-
 426 denda, and the network’s live country-profile data portal, cross-checked against
 427 IFC press releases announcing individual member accessions; economies that
 428 joined SBFN only after their WBES survey year are coded as non-adopters
 429 as of that year (Section 2). For the global sample’s remaining 157 economy-
 430 years, we draw membership status and join year directly from the SBFN data
 431 portal’s full published member roster (67 members as of the portal’s most

recent update), applying the identical ex-ante rule: an economy is coded as an adopter only if its join year precedes its survey year. Six economy-years in the global sample (Central African Republic, Cameroon, Chad, Republic of Congo, Equatorial Guinea, and Gabon, all 2023–2024) are recorded on the roster under a regional entry, “Central African States,” which we interpret as referring to the CEMAC zone’s common banking supervisor; four small-island economy-years (Antigua and Barbuda, Grenada, St. Lucia, St. Vincent and the Grenadines, all 2025) are similarly recorded under “Eastern Caribbean States,” which we interpret as the Eastern Caribbean Central Bank’s member states. Both interpretations are flagged explicitly here as inferences rather than explicit per-country listings. Regulatory capacity is measured with the World Bank’s Worldwide Governance Indicators (WGI) Regulatory Quality estimate, matched to each economy’s exact survey year (or, where not yet published, the most recent prior year available) via the World Bank DataBank API; WGI does not cover Taiwan, which is accordingly dropped from global-sample regressions.

A natural alternative to the cross-sectional design used throughout this paper would exploit SBFN’s own staggered adoption timing—join years in our sample span 2012–2024 (Section 2)—in a staggered difference-in-differences or event-study design around each economy’s own adoption date, the identification strategy this literature (and this special issue specifically) most often asks for. We do not attempt this, and it is worth being explicit about why rather than leaving the gap implicit. The WBES Green Economy module is not a repeated panel instrument: it has been fielded to a given country, in most cases, exactly once, as a special one-off module attached to a single

survey wave, and even the small subset of economies in our sample surveyed more than once by the WBES core instrument were not re-surveyed with a comparable Green Economy module in a second wave; the six supplementary economies in our own extension sample (Section 5.5) illustrate the problem directly, since even they differ from one another, and from the primary sample, in exact item wording and randomized sub-module assignment, precluding a consistently defined outcome measured twice for the same country before and after its SBFN join date. A within-country before/after comparison around each economy’s own adoption date is accordingly not implementable with a consistent outcome variable in the currently published WBES microdata, for this specific outcome construct. This is a genuine data-availability constraint on the identification strategy available to us, not a design choice; we flag it here as a first-class limitation of the paper’s identification, and return to its consequences in Section 7.

4.2. Variable construction

Outcome. Our primary dependent variable, *green practice adoption*, is a binary indicator equal to one if a firm reports having adopted at least one of seven Green Economy module practices over the preceding three years: more climate-friendly on-site energy generation, an energy-management system, waste-minimization or recycling measures, air-pollution control measures, other pollution-control measures, any energy-efficiency measure, or the use of on-site renewable energy. We also construct a continuous *green adoption index*, the mean of the same seven binary items, for robustness.

Treatment. *Bank credit access* is a binary indicator equal to one if the firm reports holding a line of credit or loan from a formal financial institution

482 at the time of the survey.

483 **Country-level moderators.** *SBFN member* is a binary indicator equal
484 to one if the economy had adopted a national SBFN sustainable-finance
485 policy framework as of the survey year. *Regulatory quality* is the WGI Reg-
486 ulatory Quality estimate for the survey year (continuous, standardized to
487 mean zero, unit standard deviation within the estimation sample for the
488 hierarchical and machine-learning specifications).

489 **Controls.** Firm size (small/medium/large, per the Enterprise Surveys’
490 own sampling strata), broad sector (manufacturing, services, construction),
491 log sales, an exporter dummy (any direct or indirect export share), and a
492 foreign-ownership dummy (any private foreign ownership share).

493 Table 1 lists every variable used in the analysis together with its exact
494 WBES question code and source, in the interest of full transparency about
495 variable construction given the multi-source nature of the merge.

496 4.3. Empirical strategy

497 We estimate the relationship between bank credit access and green prac-
498 tice adoption, and its moderation by SBFN status and regulatory quality,
499 in five stages of increasing methodological sophistication, each addressing a
500 specific limitation of the one before it.

501 **Stage 1 — classical benchmark.** We first estimate a pooled logistic
502 regression,

$$\Pr(\text{Green}_{i,c} = 1) = \Lambda\left(\beta_1 \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + \delta_{s(i)}\right) \quad (1)$$

503 for firm i in country c , where $\mathbf{X}_{i,c}$ is the firm-control vector and $\delta_{s(i)}$ is a
504 sector fixed effect, with standard errors clustered by country. We deliberately

do not include country fixed effects in this specification: because SBFN_c is constant within country in a single cross-sectional wave, a country fixed effect would absorb it—and its interaction with $\text{Credit}_{i,c}$ —completely, making β_2 and β_3 unidentifiable by construction. This is not a modelling choice we are free to make differently; it is the reason a purely classical fixed-effects approach cannot answer H2/H3 as posed, and we report this specification, extended with a triple interaction for $\text{Regulatory quality}_c$, precisely to make that limitation visible.

Stage 2 — Bayesian hierarchical (multilevel) model. We re-estimate the same relationship allowing country to enter as a random rather than fixed effect, with a random slope on Credit :

$$\text{Green}_{i,c} \sim \text{Bernoulli}(p_{i,c}) \quad (2)$$

$$\begin{aligned} \text{logit}(p_{i,c}) = & (\beta_1 + u_{1,c}) \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) \\ & + \beta_4 (\text{Credit}_{i,c} \times \text{RegQuality}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + (\alpha + u_{0,c}) \end{aligned} \quad (3)$$

$$\begin{pmatrix} u_{0,c} \\ u_{1,c} \end{pmatrix} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}) \quad (4)$$

estimated via Hamiltonian Monte Carlo (the No-U-Turn Sampler, NUTS) in PyMC. This specification is the paper’s primary vehicle for testing H2 and H3: because country enters through a variance component rather than a full set of dummies, the country-level moderators SBFN_c and RegQuality_c remain identified, while the random slope $u_{1,c}$ still absorbs unobserved country heterogeneity in the credit-green relationship and, together with partial pooling across countries, gives more defensible inference than cluster-robust standard errors computed over only 41 clusters (Cameron and Miller, 2015).

524 **Stage 3 — causal forest / double machine learning.** To avoid
 525 imposing a linear interaction functional form on how the treatment effect
 526 of credit access varies across the institutional landscape, we additionally es-
 527 timate a Causal Forest with Double Machine Learning (Wager and Athey,
 528 2018; Chernozhukov et al., 2018; Athey et al., 2019), using gradient-boosted
 529 nuisance models for the outcome and treatment propensity, and firm size, ex-
 530 port status, SBFN status, and regulatory quality as effect-modifying features.
 531 This lets us recover conditional average treatment effects (CATEs)—the esti-
 532 mated effect of credit access on green adoption, separately for SBFN members
 533 versus non-members, and separately by regulatory-quality tercile—without
 534 imposing that the moderation take the single-coefficient interaction form as-
 535 sumed in Stages 1–2. We use “causal” here in the qualified sense standard
 536 in this literature: the forest’s identifying assumption is unconfoundedness
 537 conditional on the observed covariate set (firm size, export status, SBFN
 538 status, regulatory quality), not an exogenous source of variation in credit
 539 access itself. This is exactly the same assumption underlying the logit and
 540 hierarchical specifications; Stage 3 relaxes the functional form imposed on
 541 that assumption from linear to fully flexible, but it does not relax the as-
 542 sumption, and the CATEs reported below should be read accordingly as
 543 doubly-robust conditional associations rather than causal effects identified
 544 off a natural experiment.

545 **Stage 4 — small-sample supplementary check.** We additionally
 546 re-estimate a reduced-form version of the credit-green relationship using a
 547 waste-minimization-adoption outcome, available in four economies (India
 548 2022, Indonesia 2023, Peru 2023, Timor-Leste 2021) whose Green Economy

modules asked this item under directly comparable wording. This outcome is not part of the standardized cross-country database underlying Stage 5 below, so it offers genuinely independent, if small-sample, supplementary evidence.

Stage 5 — global-sample replication with country fixed effects.

Finally, and centrally, we re-estimate the full specification on the 162-economy global sample described in Section 4, using CO2-emissions monitoring as the outcome. Because this sample spans far more country clusters than the primary sample, we can now afford a country-fixed-effects specification: δ_c absorbs SBFN_{*c*} and RegQuality_{*c*}’s main effects (collinear with the fixed effect by construction, since both are constant within country-year) while leaving their firm-level interactions with Credit_{*i,c*} fully identified, since credit access itself varies within country. This is a materially sharper test of H2/H3 than Stage 1’s clustered-SE approach, on an entirely independent sample and outcome from Stages 1–3; we report it alongside a no-fixed-effects specification for direct comparability with the primary sample’s Table 3.

5. Results

5.1. Descriptive patterns

Table 2: Sample composition, by economy.

Economy	<i>N</i> firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Albania 2019	377	2019	0	0.09	0.609	0.407
Armenia 2020	546	2020	0	0.21	0.684	0.482

Economy	N firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Azerbaijan 2019	225	2019	0	-0.10	0.418	0.180
Belarus 2018	600	2018	0	-0.79	0.662	0.425
Bosnia and Herze- govina 2019	362	2019	0	-0.38	0.581	0.565
Bulgaria 2019	772	2019	0	0.46	0.604	0.467
Croatia 2019	404	2019	0	0.38	0.708	0.448
Cyprus 2019	240	2019	0	1.04	0.746	0.711
Czechia 2019	502	2019	0	1.12	0.725	0.526
Egypt 2020	3,075	2020	1	-0.64	0.303	0.065
Estonia 2019	360	2019	0	1.54	0.738	0.412
Georgia 2019	581	2019	1	0.88	0.441	0.501
Greece 2018	600	2018	0	0.29	0.825	0.499
Hungary 2019	805	2019	0	0.46	0.673	0.537
Italy 2019	760	2019	0	0.70	0.506	0.230
Jordan 2019	601	2019	1	0.16	0.471	0.248
Kazakhstan 2019	1,446	2019	0	0.11	0.519	0.257
Kosovo 2019	271	2019	0	-0.38	0.698	0.425
Kyrgyz Republic 2019	360	2019	1	-0.42	0.606	0.299
Latvia 2019	359	2019	0	1.00	0.824	0.386
Lebanon 2019	532	2019	0	-0.56	0.452	0.365
Lithuania 2019	358	2019	0	0.98	0.678	0.342
Malta 2019	242	2019	0	0.83	0.901	0.479
Moldova 2019	360	2019	0	-0.11	0.619	0.328

Economy	<i>N</i> firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Mongolia 2019	360	2019	1	-0.07	0.625	0.492
Montenegro 2019	150	2019	0	0.29	0.387	0.576
Morocco 2019	1,096	2019	1	0.03	0.478	0.251
North Macedonia 2019	360	2019	0	0.19	0.572	0.508
Poland 2019	1,369	2019	0	0.83	0.569	0.403
Portugal 2019	1,062	2019	0	1.00	0.812	0.600
Romania 2019	814	2019	0	0.36	0.438	0.474
Russia 2019	1,323	2019	0	-0.35	0.503	0.310
Serbia 2019	361	2019	0	0.06	0.654	0.585
Slovak Republic 2019	429	2019	0	0.73	0.782	0.386
Slovenia 2019	409	2019	0	0.88	0.793	0.674
Tajikistan 2019	352	2019	0	-0.80	0.523	0.194
Tunisia 2020	615	2020	1	-0.11	0.391	0.413
Türkiye 2019	1,663	2019	1	-0.02	0.346	0.345
Ukraine 2019	1,337	2019	0	-0.29	0.715	0.244
Uzbekistan 2019	1,239	2019	0	-0.92	0.751	0.240
West Bank and Gaza 2019	365	2019	0	-0.15	0.428	0.178

Note: Reg. qual. is the WGI Regulatory Quality estimate for the survey year. Green rate is the unconditional share of firms reporting adoption of at least one Green Economy practice. Credit rate is the share of firms reporting a bank credit line or loan.

Table 1: Variable definitions and sources.

Variable	Definition	WBES code	Source
green_adoption_binary	=1 if firm adopted ≥ 1 of 7 Green Economy practices in last 3 years	BMGc23b, c23d, c23e, c23f, c23j, c25, e5	WBES Green Economy Module (2018–2020)
fin_has_credit_line	=1 if firm has a line of credit or loan	K8	WBES Access to Finance Module
sbfn_member	=1 if SBFN policy adopted as of survey year	n/a	SBFN Global Progress Reports; data.sbfnetwork.org
wgi_regulatory_quality	WGI Regulatory Quality estimate, survey year	n/a	World Bank DataBank API (GOV_WGI_RQ_EST)
size_cat	Firm size stratum (Small/Medium/Large)	A6A	WBES Sampling Frame
sector_broad	Broad sector (Manufacturing/Services/Construction)	A4AX	WBES Sampling Frame
log_sales	Log of total annual sales	D2	WBES General Information Module
foreign_owned_dummy	=1 if any foreign ownership share $> 0\%$	B2B	WBES Ownership Module
exporter_dummy	=1 if any export sales share $> 0\%$	D3B, D3C	WBES Sales Module

567 Table 2 reports the full 41-economy sample composition; firm counts range
 568 from 150 (Montenegro) to 3,075 (Egypt), and the raw, unconditional green-
 569 practice-adoption rate ranges more than three-fold across economies, from
 570 30.3% (Egypt) to 90.1% (Malta). Figure 1 maps this variation alongside
 571 SBFN policy status: 8 of the 41 primary-sample economies had adopted
 572 an SBFN framework as of their survey year (Georgia, Egypt, Jordan, Kyr-
 573 gyz Republic, Mongolia, Morocco, Tunisia, and Türkiye)—13 of the full 47-
 574 economy sample once the five (of six) SBFN-member small-sample supple-
 575 mentary economies are counted—and every European Union member state
 576 in the sample is, consistent with SBFN’s mandate, a non-adopter. At the
 577 firm level, bank credit access correlates positively with the green-adoption
 578 index ($r = 0.18$) but is essentially uncorrelated with a firm’s own perceived
 579 finance obstacle ($r = -0.02$)—a first hint, well before any regression, that re-
 580 alized access to credit rather than perceived financing difficulty is what tracks
 581 green behaviour, and the two are evidently not interchangeable measures of
 582 the same underlying constraint.

583 Figure 2 previews the paper’s central finding graphically before a sin-
 584 gle regression is estimated: plotting each economy’s green-adoption rate
 585 against its regulatory-quality score and fitting separate lines for SBFN mem-
 586 bers and non-members produces two lines of very similar, modestly positive
 587 slope, separated by a roughly 15–20 percentage-point vertical gap—SBFN-
 588 member economies show systematically *lower* raw green-adoption rates at
 589 every level of regulatory quality, not a steeper relationship between regu-
 590 latory quality and adoption. That combination—a level difference with no
 591 slope difference—is exactly the signature the regression evidence below con-

592 firms formally.

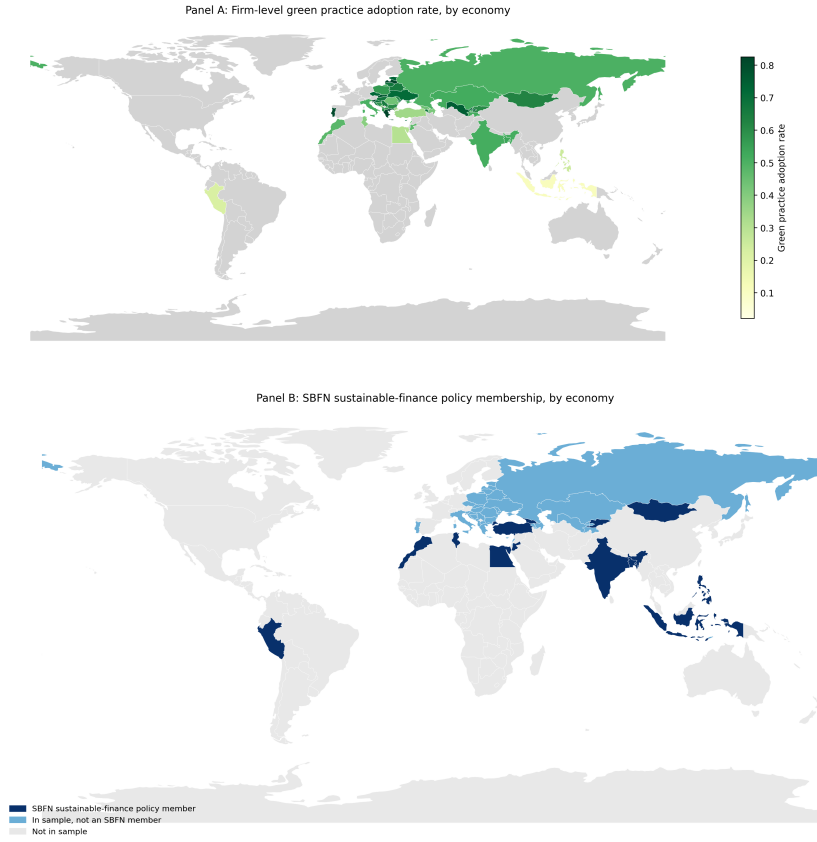


Figure 1: Green practice adoption and SBFN policy membership, by economy.

593 5.2. Baseline classical benchmark

594 Table 3 reports the pooled logistic baseline. Column M1 confirms H1 on
 595 its own terms: firms with a bank credit line are significantly more likely to
 596 have adopted a green practice ($b = 0.571$, $p < 0.001$; average marginal effect
 597 $= 12.95$ percentage points, 95% CI $[7.7, 18.2]$)—holding a formal credit line is
 598 associated with roughly the same increase in the probability of green adoption
 599 as moving from a small to a large firm, and a somewhat larger increase than

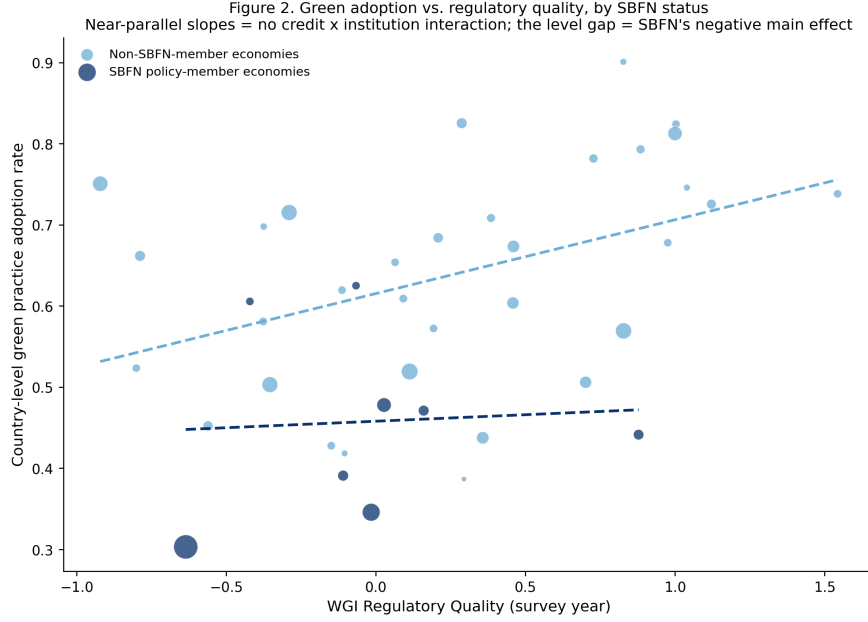


Figure 2: Green adoption vs. regulatory quality, by SBFN status.

being an exporter. Column M2 adds SBFN status and its interaction with credit access: the SBFN main effect is large, negative, and precisely estimated ($b = -0.991$, $p < 0.001$), while the credit $\times$ SBFN interaction is small and statistically indistinguishable from zero ($b = 0.049$, $p = 0.773$). Column M3's fully saturated triple interaction with regulatory quality tells the same story at greater length: every interaction term involving credit access is small and insignificant, while regulatory quality's uninteracted relationship with adoption is positive, if only marginally significant in this particular fully-saturated specification ($b = 0.278$, $p = 0.110$).

As flagged in Section 4, this specification cannot include country fixed effects without mechanically absorbing the very SBFN and regulatory-quality terms the hypotheses concern, so what Table 3 shows is a genuine test of

H2/H3, not a placeholder—and that test returns no interaction. The question the remaining two stages are built to answer is whether that null result reflects a real absence of policy moderation, or an artefact of forcing a two-level question through a single-level model with country-clustered standard errors computed over only 41 clusters, a setting in which such standard errors are known to be unreliable (Cameron and Miller, 2015).

5.3. Bayesian hierarchical model — primary test of H2/H3

The hierarchical specification—country-varying random slopes on credit access, cross-level interactions with SBFN status and standardized regulatory quality, four chains of 1,000 post-warmup draws each, all $\hat{R} \leq 1.01$ and ess_{bulk} between 797 and 2,270—is built specifically to give H2 and H3 a fair hearing that the absorbed classical specification could not. It does not change the conclusion. Table 4 reports the full posterior summary; three results stand out.

First, the firm-level credit effect (H1) survives unchanged in substance: posterior mean 0.33 (89% equal-tailed interval [0.18, 0.49]), comfortably excluding zero. Second, both country-level institutional variables retain clean, precisely estimated *main* effects on the baseline adoption rate—SBFN membership net-negative (-0.57, 89% ETI [-1.00, -0.09]), regulatory quality net-positive (0.34, 89% ETI [0.16, 0.53])—reproducing, with proper hierarchical uncertainty quantification rather than an absorbed or clustered approximation, exactly the level-shift pattern Figure 2 shows visually. Third, and centrally for H2 and H3, neither cross-level interaction is distinguishable from zero: credit $\times$ SBFN, 0.072 (89% ETI [-0.26, 0.40]); credit $\times$ regulatory quality (standardized), 0.004 (89% ETI [-0.13, 0.14])—a near-exact zero on

Table 3: Baseline classical logit regressions (dependent variable: green practice adoption, binary). Country-clustered standard errors in parentheses.

	M1	M2	M3
Credit access	0.571*** (0.123)	0.459*** (0.090)	0.373*** (0.085)
SBFN member		-0.991*** (0.178)	-0.844*** (0.209)
Credit $\times$ SBFN		0.049 (0.168)	0.019 (0.140)
Regulatory quality			0.278 (0.174)
Credit $\times$ Reg. quality			0.144 (0.142)
SBFN $\times$ Reg. quality			0.053 (0.261)
Credit $\times$ SBFN $\times$ Reg. quality			-0.104 (0.272)
Log sales	0.061 (0.039)	0.041 (0.034)	0.068** (0.032)
Foreign owned	0.219 (0.143)	0.244* (0.139)	0.220 (0.143)
Exporter	0.537*** (0.092)	0.471*** (0.074)	0.412*** (0.079)
Sector, size controls	Yes	Yes	Yes
N	23,914	23,914	23,914
Pseudo R^2	0.058	0.089	0.093

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

637 the latter, not merely an imprecise one.

638 We read this as an honest and, we think, informative finding rather than
639 a disappointing one: giving the moderation hypothesis a hierarchical model
640 built for precisely this two-level question still returns no interaction, while
641 sharpening (relative to the pooled baseline) the confidence with which we
642 can state that SBFN status and regulatory quality shift the *level* of green
643 adoption directly.

Table 4: Bayesian hierarchical model, posterior summary (four chains, 1,000 post-warmup draws each).

Parameter	Mean	SD	89% ETI low	89% ETI high	$\hat{R}$
Intercept	-1.93	0.246	-2.30	-1.50	1.00
Credit access	0.33	0.097	0.18	0.49	1.00
SBFN member	-0.57	0.300	-1.00	-0.09	1.01
Credit $\times$ SBFN	0.072	0.211	-0.26	0.40	1.00
Reg. quality (std.)	0.344	0.116	0.16	0.53	1.00
Credit $\times$ Reg. quality	0.004	0.087	-0.13	0.14	1.00

644 5.4. Causal forest / heterogeneous treatment effects

645 Table 5 and Figure 5 report the Causal Forest DML estimates, which im-
646 pose no linear-interaction functional form and let the treatment effect vary
647 flexibly across firm and country covariates. The overall average treatment
648 effect of credit access is 0.073 (95% CI [-0.040, 0.186])—directionally consis-
649 tent with, though smaller in magnitude and less precisely estimated than,
650 the logit average marginal effect of 0.130 in Section 5.2, reflecting the more

651 conservative variance properties of a doubly-robust, cross-fitted nonparamet-
 652 ric estimator relative to a parametric logit on the same data. Critically,
 653 the conditional average treatment effects triangulate with Section 5.3 rather
 654 than contradicting it: the estimated effect is 0.072 for firms in non-SBFN
 655 economies versus 0.076 for firms in SBFN economies, and 0.075 / 0.070 /
 656 0.075 across the low/middle/high regulatory-quality terciles respectively—
 657 three different institutional cuts, three essentially flat effects, all visibly over-
 658 lapping in Figure 5. Two entirely independent estimators, one parametric
 659 and hierarchical, one nonparametric and forest-based, agree on the absence
 660 of institutional moderation.

661 The forest’s feature-importance decomposition adds a genuinely new re-
 662 sult rather than only confirming the null, and speaks directly to the auxiliary
 663 H4 introduced in Section 3.4: of the four candidate effect-modifiers, firm size
 664 (log sales) accounts for 75.6% of the model’s explained heterogeneity, versus
 665 19.9% for regulatory quality, 2.6% for SBFN status, and 1.9% for exporter
 666 status. Whatever heterogeneity exists in how strongly credit access pre-
 667 dicts green adoption is, on this evidence, primarily a firm-level rather than a
 668 country-institutional phenomenon, supporting H4. Collapsing this into the
 669 Enterprise Surveys’ own coarse size strata, however, does not reveal a clean
 670 monotonic gradient: the estimated effect is 0.080 for small firms, 0.075 for
 671 medium firms, and 0.057 for large firms (Table 6; all with wide, overlap-
 672 ping confidence intervals spanning zero), which if anything runs opposite to
 673 a simple “larger firms better absorb a green investment’s fixed costs” story
 674 rather than confirming it. Read alongside the 75.6% importance share for the
 675 continuous size measure, we take this as evidence that the heterogeneity the

676 forest is detecting is more granular or non-monotonic across the firm-size dis-
677 tribution than a three-category collapse can reveal, rather than evidence for
678 any particular direction of a size gradient—a qualification we report explicitly
679 rather than paper over with a tidier-sounding but unsupported claim. This
680 redirects, rather than closes, the paper’s institutional-moderation question,
681 and we return to it in Section 6.

Table 5: Causal forest DML: average treatment effect, CATEs, and feature importances.

Estimate	Value	95% CI low	95% CI high
Overall ATE	0.0733	-0.0396	0.1862
CATE SBFN non-member	0.0723	-0.0300	0.1746
CATE SBFN member	0.0758	-0.0591	0.2107
CATE reg. quality tercile: low	0.0753	-0.0568	0.2075
CATE reg. quality tercile: mid	0.0697	-0.0355	0.1749
CATE reg. quality tercile: high	0.0749	-0.0203	0.1701
Feature	Importance		
SBFN member	0.0257		
Regulatory quality	0.1993		
Log sales	0.7562		
Exporter dummy	0.0189		

682 5.5. Small-sample supplementary check: waste-minimization adoption

683 Before turning to the global-sample replication that is this paper’s second
684 major test (Section 5.7), Table 7 reports a smaller supplementary check using
685 waste-minimization adoption — an outcome not captured in the standardized

Table 6: Causal forest CATEs by firm-size category.

Size category	CATE	95% CI low	95% CI high
Small	0.0797	-0.0409	0.2004
Medium	0.0747	-0.0344	0.1838
Large	0.0569	-0.0428	0.1565

cross-country database underlying the global sample, and so a genuinely independent (if small-sample) data point — available in four economies whose Green Economy modules asked this item under directly comparable wording: India 2022, Indonesia 2023, Peru 2023, and Timor-Leste 2021. Three of these four are themselves SBFN members at various progression stages, so this sample cannot re-test the SBFN-adoption contrast; what it can test is whether the credit-to-green channel documented in Sections 5.2–5.4 is a peculiarity of the ECA-MENA sample or generalizes elsewhere. It generalizes: pooled across the four economies, having an overdraft facility is associated with significantly higher odds of waste-minimization adoption ($b = 0.422$, $p < 0.001$, $N = 9,038$, 4 countries). The direction of the core finance-green relationship replicates on a different continent, a different half-decade, and a narrower outcome definition than the one anchoring Sections 5.2–5.4 — a preview, on four economies, of the much larger replication in Section 5.7.

5.6. Additional robustness

Table 9 subjects the primary-sample finding to four further checks, all retaining sector fixed effects, firm controls, and country-clustered standard errors (no country fixed effects, for the reason given in Section 4). First,

Table 7: Small-sample supplementary check: descriptive rates.

Economy	N	Waste-min. rate	Overdraft rate	SBFN member
India 2022	9,376	0.515	0.621	1
Indonesia 2023	2,955	0.336	0.144	1
Peru 2023	987	0.806	0.549	1
Timor-Leste 2021	238	0.061	0.089	0

replacing the binary outcome with the continuous seven-item green-adoption index and estimating by OLS reproduces the same pattern to the decimal: credit access positive and significant ($b = 0.053$, $p < 0.001$), SBFN main effect negative and significant ($b = -0.092$, $p < 0.001$), interaction indistinguishable from zero ($b = 0.006$, $p = 0.739$).

Second, replacing bank credit access with two alternative finance measures produces one genuine point of nuance worth reporting rather than smoothing over. Using overdraft-facility access instead of a credit line/loan, the main overdraft effect remains positive and significant ($b = 0.274$, $p = 0.011$), the SBFN main effect remains negative and significant ($b = -0.905$, $p < 0.001$)—but the overdraft $\times$ SBFN interaction is now negative and significant ($b = -0.346$, $p = 0.031$), the one interaction term in the entire paper that clears a conventional significance threshold, and it runs opposite to the direction H2 predicts. Using the reverse-coded perceived finance-obstacle measure in place of realized access, by contrast, both the main effect and its SBFN interaction are precisely zero ($b = -0.010$, $p = 0.827$; interaction $b = -0.012$, $p = 0.849$)—consistent with the descriptive correlation noted in Section 5.1. We read the overdraft result as suggestive rather than conclusive given it is one significant coefficient among more than a dozen

723 interaction tests across the paper, but it is at least directionally consistent
 724 with a substantive story worth flagging for future work: if SBFN-guided
 725 lending is channelled preferentially through the term loans and credit lines
 726 that finance long-horizon capital expenditure, rather than through short-term
 727 overdraft facilities, we might expect exactly this pattern—a null-to-positive
 728 interaction on credit lines and a negative one on overdrafts—though our
 729 data cannot directly test whether SBFN-guided credit is disproportionately
 730 term-structured.

731 Third, splitting the sample into manufacturing ($N = 13,340$) and services
 732 ($N = 9,538$) subsamples shows the credit-access main effect, the SBFN main
 733 effect, and the null credit $\times$ SBFN interaction all replicate within each sector
 734 separately (manufacturing: credit $b = 0.480$, $p < 0.001$, interaction $b =$
 735 -0.040 , $p = 0.826$; services: credit $b = 0.448$, $p < 0.001$, interaction $b =$
 736 0.195 , $p = 0.426$)—the paper’s central null is not an artefact of pooling two
 737 structurally different production technologies.

738 Because the overdraft interaction (the one coefficient in the paper that
 739 clears a conventional significance threshold) is one of many credit $\times$ institutional-
 740 moderator tests reported across the frequentist specifications in this paper,
 741 we additionally apply a Benjamini–Hochberg false-discovery-rate correction
 742 across the full family of 13 such coefficients reported under a classical (non-
 743 Bayesian, non-forest) estimator in Tables 3, 9, and 10 (Table 8; script and
 744 output in the replication package). The overdraft interaction’s raw $p = 0.031$
 745 becomes $p = 0.403$ after correction, and no coefficient in the family survives
 746 at a false discovery rate of 0.10 or even 0.40. We read this as reinforcing, with
 747 a formal multiple-testing accounting rather than only a qualitative caveat,

748 the suggestive-not-conclusive framing we already give this result in Section 6:
749 an isolated nominally-significant interaction among more than a dozen tested
750 is exactly what a pure false-discovery process would be expected to produce
751 roughly one time in twenty, and this correction does not let us reject that
752 account.

Table 8: Multiple-testing correction: Benjamini–Hochberg false discovery rate applied to every credit $\times$ institutional-moderator interaction coefficient reported under a frequentist estimator (Tables 3, 9, 10). Table 3 model M3 entries marked $\dagger$ are approximate p -values recovered from the reported coefficient and standard error via a normal approximation ($z = \text{coef}/\text{se}$), since the text reports only the coefficient and standard error for these terms.

Source	Term	Raw p	BH-adjusted p
Table 9 (b1) Overdraft	Overdraft $\times$ SBFN	0.031	0.403
Table 10 M2 (country FE)	Credit $\times$ SBFN	0.205	0.799
Table 10 M1 (no country FE)	Credit $\times$ Reg. quality †	0.301	0.799
Table 3 M3 †	Credit $\times$ Reg. quality	0.310	0.799
Table 10 M2 (country FE)	Credit $\times$ Reg. quality	0.346	0.799
Table 9 (c2) Services	Credit $\times$ SBFN	0.426	0.799
Table 10 M1 (no country FE)	Credit $\times$ SBFN	0.430	0.799
Table 3 M3 †	Credit $\times$ SBFN $\times$ Reg. quality	0.702	0.892
Table 9 (a) Continuous index, OLS	Credit $\times$ SBFN	0.739	0.892
Table 3 M2	Credit $\times$ SBFN	0.773	0.892
Table 9 (c1) Manufacturing	Credit $\times$ SBFN	0.826	0.892
Table 9 (b2) Finance obstacle (reverse-coded)	Obstacle $\times$ SBFN	0.849	0.892
Table 3 M3 †	Credit $\times$ SBFN	0.892	0.892

Table 9: Additional robustness checks.

Specification	Finance coef. (p)	SBFN coef. (p)	Interaction coef. (p)
(a) Continuous index, OLS	0.053 (0.000)	-0.092 (0.000)	0.006 (0.739)
(b1) Overdraft	0.274 (0.011)	-0.905 (0.000)	-0.346 (0.031)
(b2) Finance obstacle (reverse-coded)	-0.010 (0.827)	-1.011 (0.000)	-0.012 (0.849)
(c1) Manufacturing subsample	0.480 (0.000)	-1.047 (0.000)	-0.040 (0.826)
(c2) Services subsample	0.448 (0.000)	-0.914 (0.001)	0.195 (0.426)

753 5.7. Global-sample replication with country fixed effects

754 The preceding five subsections establish the paper’s central finding on the
755 primary sample. We now subject that finding to what we consider the most
756 demanding test available to us: complete re-estimation on an independently
757 assembled, 162-economy global sample (Section 4), using a different outcome
758 variable (CO2-emissions monitoring rather than the seven-item composite),
759 different survey years (2021–2026 rather than 2018–2020), and, for the first
760 time in the paper, a country-fixed-effects specification that this much larger
761 set of clusters can support. Figure 3 maps the combined reach of the two
762 samples together: 166 distinct economies, spanning every region and, through
763 the global sample specifically, extending for the first time in this literature
764 into high-income economies with no SBFN involvement at all.

765 Table 10 reports two specifications. Column M1, without country fixed
766 effects, is the direct global-sample analogue of Table 3’s primary-sample base-
767 line: credit access remains positive and significant ($b = 0.198$, $p = 0.039$,
768 $N = 89,797$, 159 countries), while the SBFN main effect ($b = -0.171$,
769 $p = 0.353$), the credit $\times$ SBFN interaction ($b = 0.278$, $p = 0.430$), and
770 every regulatory-quality interaction are statistically indistinguishable from

771 zero. Column M2 imposes country fixed effects, absorbing the SBFN and
772 regulatory-quality main effects by construction (they are collinear with δ_c ,
773 since both are constant within a given country-year) while leaving their firm-
774 level interactions with credit access fully identified, because credit access
775 itself varies within country. This is a sharper test than anything the primary
776 sample’s single 2018–2020 wave can support, and it returns the same answer
777 with even more precision: credit access is strongly significant ($b = 0.249$,
778 $p < 0.001$), while the credit $\times$ SBFN interaction ($b = 0.506$, $p = 0.205$) and
779 the credit $\times$ regulatory-quality interaction ($b = -0.044$, $p = 0.346$) remain
780 null.

781 We regard this as the paper’s most important robustness result. It is not
782 a re-analysis of the same data under a different label: it is a different outcome
783 construct, a different set of economies (only a handful of which overlap with
784 the primary sample), a different half-decade, and an econometric specification
785 — country fixed effects — that is only available at all because this second
786 sample has enough clusters to support it. That the qualitative conclusion
787 is identical across both samples and across the full set of four estimators
788 reported in this paper (classical logit, Bayesian hierarchical, causal forest, and
789 now country-fixed-effects logit) is, we think, considerably stronger evidence
790 for the paper’s central claim than either sample could offer alone.

791 *5.8. Results summary across all specifications*

792 Sections 5.2–5.7 report ten tables spanning four estimators and two inde-
793 pendent samples. Figure 4 consolidates every coefficient bearing on H1–H3
794 into a single view. Panel A shows the credit-access effect (H1): every point
795 estimate is positive and every confidence interval excludes zero, in both sam-

Table 10: Global-sample regressions (dependent variable: CO2-emissions monitoring, binary). Country-clustered standard errors in parentheses; M2's SBFN and regulatory-quality main effects are absorbed by country fixed effects and are not separately estimable.

	M1: no country FE	M2: country FE
Credit access	0.198** (0.096)	0.249*** (0.057)
SBFN member	-0.171 (0.184)	— (absorbed)
Credit $\times$ SBFN	0.278 (0.352)	0.506 (0.399)
Regulatory quality (std.)	0.155 (0.096)	— (absorbed)
Credit $\times$ Reg. quality	0.092 (0.089)	-0.044 (0.047)
Log sales	0.127*** (0.027)	0.288*** (0.028)
Foreign owned	0.621*** (0.083)	0.528*** (0.046)
Exporter	0.456*** (0.094)	0.335*** (0.044)
Country fixed effects	No	Yes
N	89,797	89,797
Countries	159	159

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Taiwan dropped (no WGI coverage).

Figure 3. Combined sample coverage: 166 economies (41 primary rich-module + 162 global minimal-indicator, net of overlap)

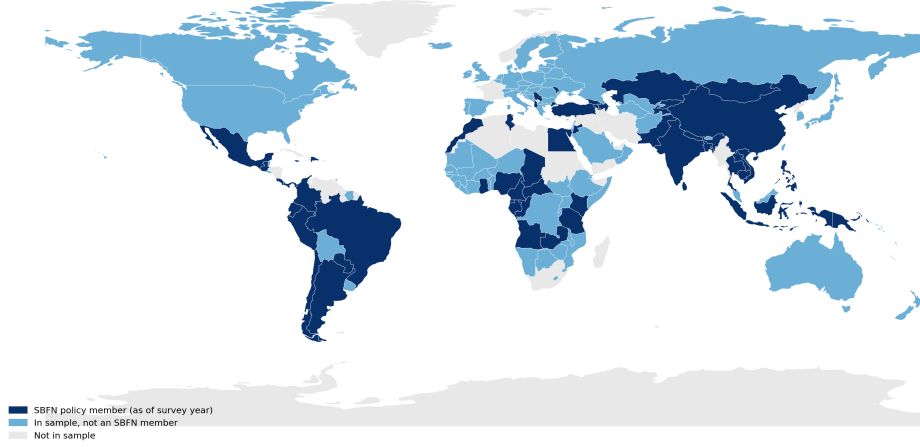


Figure 3: Combined sample coverage: 166 economies (41 primary + 162 global, net of overlap), shaded by SBFN status.

796 ples and all four estimators. Panels B and C show the credit $\times$ SBFN and
797 credit $\times$ regulatory-quality interactions (H2, H3): every single confidence in-
798 terval, across both samples and every estimator, straddles zero. Figure 5
799 shows the same non-result from a completely different angle—the causal
800 forest’s conditional average treatment effects broken out by SBFN status,
801 regulatory-quality tercile, and firm-size category all overlap both each other
802 and the overall average treatment effect. No subgroup split, by any variable
803 examined in this paper, recovers a distinguishable effect.

804 6. Discussion

805 Put in plain economic terms, the paper’s central finding is this: a firm
806 holding a bank credit line is associated with an adoption probability roughly
807 13 percentage points higher than an otherwise-similar firm without one—

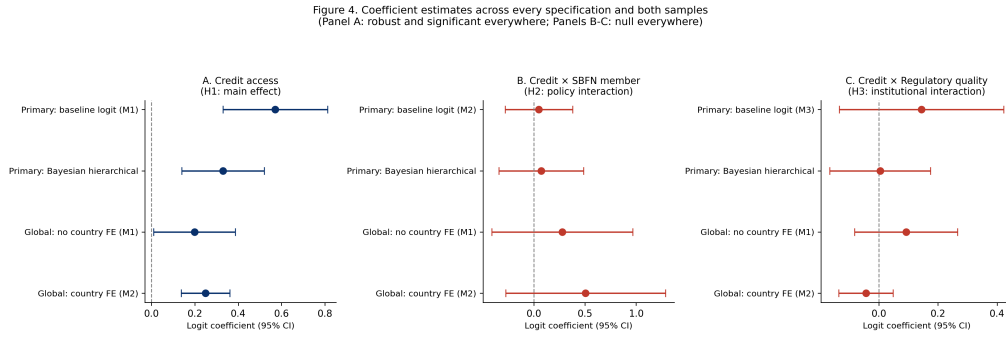


Figure 4: Coefficient estimates across every specification and both samples.

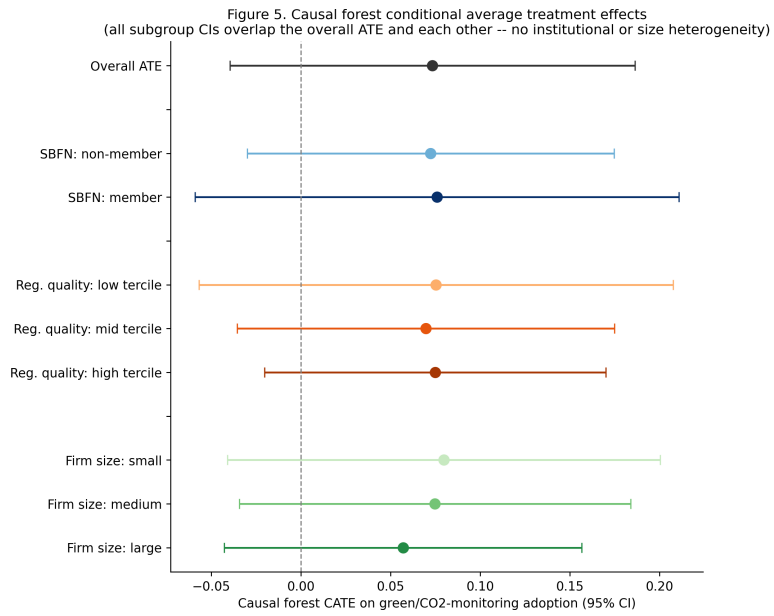


Figure 5: Causal forest conditional average treatment effects, by subgroup.

an association net of the observed firm and country controls, not a causally identified effect, for the reasons given in Limitation 1 below—comparable in magnitude to the gap between a small firm and a large one—and that gap is essentially the same size whether the firm sits in an SBFN sustainable-finance policy adopter or a non-adopter, and essentially the same size whether the country’s regulatory quality sits in the bottom, middle, or top third of the sample (Figure 4). Firms in SBFN-adopting economies are, on average, some 15–20 percentage points *less* likely to have adopted a green practice at all—a level gap, not a slope gap.

This convergence matters most because of how differently the two samples could have failed to agree. The primary and global samples share almost no economies in common, were fielded in different half-decades, measure green practice adoption with entirely different survey items, and reach institutional settings—high-income, non-SBFN economies such as Australia and Germany—that the primary sample cannot touch at all. A finding that held only in the 41-country ECA-MENA sample could plausibly be a regional or period-specific artefact; a finding that survives unchanged when re-estimated on 162 largely non-overlapping economies under a sharper identification strategy is considerably harder to explain away as such.

This is worth dwelling on precisely because it complicates rather than confirms the implicit premise of much of the existing green-credit-policy literature, which is disproportionately built on evidence from China’s unusually mature, unusually well-resourced green credit guidelines (Li et al., 2023; Huang et al., 2023; Dai et al., 2025). Our sample includes economies at every stage of the SBFN’s own progression framework—from Jordan and the

833 Kyrgyz Republic’s initial “Commitment” stage to Mongolia’s “Advancing” im-
834 plementation a full seven years after founding membership—and across that
835 entire range, we find no evidence that formal policy adoption changes how a
836 firm’s credit access translates into green investment behaviour. Section 5.4’s
837 most novel result gives one plausible redirection of the question, though not
838 a tidy answer to it: the heterogeneity that *does* exist in the credit-green
839 relationship is overwhelmingly a firm-size phenomenon (75.6% of explained
840 variance) rather than a country-institutional one (regulatory quality 19.9%,
841 SBFN status a mere 2.6%)—yet, as Section 5.4 reports honestly, collaps-
842 ing this into the Surveys’ own coarse Small/Medium/Large strata does not
843 recover a clean gradient in either direction, so we cannot yet say whether
844 it is small or large firms for whom the credit-green channel runs stronger,
845 only that firm scale—in some more granular or non-monotonic form than a
846 three-category split reveals—is where the forest locates most of the effect het-
847 erogeneity a country-level policy variable does not. What this does support is
848 a reorientation of where policy attention belongs: if the operative constraint
849 sits at the level of which individual firms can absorb a green investment’s
850 fixed costs once financed, rather than at the level of whether a supervisory
851 guideline nudges banks to lend more broadly, then a country-level supervisory
852 mandate operating on the extensive margin of bank lending decisions may be
853 aimed at the wrong margin regardless of its own institutional credibility—a
854 question future work with a finer-grained firm-size or capacity measure is
855 better placed to resolve than the coarse categories available here.

856 It is worth being precise about what this null does and does not establish.
857 “No interaction detected” and “no institutional complementarity exists” are

not the same claim, and we do not want the paper’s repeated confirmation of the former to be read as having settled the latter. At least three distinct readings remain open. First, our preferred reading throughout the paper: the complementarity genuinely does not operate through this channel. Second, a linear interaction term may simply be the wrong functional form to detect a genuinely threshold-shaped complementarity—SBFN’s own Multi-Tiered Progress Framework (Section 2) treats policy adoption as a graded process from an initial “Commitment” stage through “Consolidating,” and our binary membership coding (Limitation 2 below) could easily average over a complementarity that only activates once a jurisdiction clears a particular progression stage, in a way four estimators built around a single membership dummy cannot recover. Third, the Worldwide Governance Indicators’ broad, economy-wide regulatory-quality measure may simply not proxy the specific banking-supervisory capacity a green-lending guideline actually depends on (Limitation 5); a null on this proxy is evidence about the proxy’s channel, not necessarily about supervisory capacity in general. We think the first reading is the best-supported of the three given the evidence in Section 5, but we hold it as the best current account rather than as a claim the paper has definitively ruled the other two out.

A further alternative worth engaging directly, since it is the more theoretically interesting institutional-mechanism story a system-change-oriented reading of our results would want addressed: SBFN guidelines could, in principle, be working exactly as intended on the *extensive* margin of who obtains credit at all, rather than on the *intensive* margin our interaction terms test. If a green-banking guideline induces banks in adopting jurisdictions to extend

883 credit more broadly to environmentally promising firms that would otherwise
884 have gone unfinanced, that effect shows up as a shift in *which* firms hold a
885 credit line, not as a different adoption gap between financed and unfinanced
886 firms conditional on already holding one—exactly the quantity our credit $\times$
887 SBFN interaction estimates. Testing this directly would require modelling
888 credit access itself, rather than green adoption conditional on credit access,
889 as a function of SBFN status and firm environmental characteristics; our
890 design, built around the demand-side finance-to-green channel, does not do
891 this, and we flag it explicitly as an unaddressed alternative mechanism rather
892 than one our null result rules out.

893 We should also be candid about a limit of our outcome measure itself.
894 Every green-practice-adoption item in this paper is firm-reported, and we
895 cannot rule out that self-reported adoption rates differ systematically from
896 banks’ own compliance reporting to their SBFN supervisor in ways our
897 data cannot observe—a general concern about compliance-reporting-versus-
898 practice gaps that applies to any policy evaluated through the regulated
899 party’s own attestations, and one a firm-level survey instrument, however
900 carefully harmonized, cannot fully resolve on its own.

901 To our knowledge, this paper is also the first to test the SBFN frame-
902 work’s own theory of change against firm-level, rather than bank-level or
903 policy-stage, outcomes: the network’s periodic Global Progress Reports (Sec-
904 tion 2) measure and publicize each member jurisdiction’s regulatory-output
905 stage, but not whether that output changes what happens on the ground
906 for the firms the policy ultimately targets. Read this way, our null result is
907 directly relevant to the SBFN’s and the IFC’s own self-evaluation practice, in-

908 dependent of what it implies for the academic institutional-complementarity
909 literature: a framework can progress steadily through Commitment, Develop-
910 ing, and Advancing stages by every measure the network itself tracks, while
911 the firm-level channel that is its ultimate point remains, on this evidence,
912 unmoved.

913 The one interaction that does clear a conventional significance threshold—
914 a *negative* credit $\times$ SBFN effect when finance access is measured through
915 overdraft facilities rather than term credit lines (Section 5)—is, we think,
916 more consistent with this redirection than with a simple failure of green
917 banking policy. Overdrafts finance working capital, not the kind of multi-year
918 capital expenditure our outcome variable measures; if SBFN-guided lending
919 is disproportionately channelled through the term-loan instruments better
920 suited to financing a solar installation or an energy-efficiency retrofit, exactly
921 the pattern we find—a null-to-positive effect on credit lines, a negative one
922 on overdrafts—is what a supervisor successfully steering credit toward long-
923 horizon green investment, rather than short-horizon working capital, would
924 produce. We flag this as suggestive rather than established, since it is a
925 single significant coefficient among the many interaction tests reported across
926 Tables 3–9—and, as Section 5 reports formally (Table 8), it does not survive a
927 Benjamini–Hochberg correction for multiple comparisons at any conventional
928 false discovery rate—and our data cannot directly verify the underlying loan-
929 tenor mechanism.

930 Read against this special issue’s organizing themes of state capacity and
931 policy credibility, and holding to the same best-current-account framing as
932 above rather than treating it as settled, our results sit closer to a state-

933 capacity-is-necessary-but-firm-level-frictions-bind-independently story than to
934 a straightforward institutional-complementarity one in which stronger regu-
935 latory quality unlocks a latent policy effect that would otherwise lie dor-
936 mant. Regulatory quality matters—it raises the general baseline propensity
937 to adopt green practices regardless of a firm’s own financing arrangement—
938 but it does not appear to be the missing ingredient that would let a green
939 banking guideline reshape the credit-to-green channel specifically. This sug-
940 gests, tentatively, that the binding constraint on green credit policy’s effec-
941 tiveness in our sample is less about whether the supervising jurisdiction has
942 the state capacity to make the guideline credible, and more about whether
943 firms below a certain scale can absorb a green investment’s fixed costs at all
944 once credit is available—a question about firm-level absorptive capacity that
945 sits one level down from the state-capacity question the SBFN framework
946 itself is designed to address.

947 **7. Conclusion**

948 This paper set out to test whether the wave of green banking policy
949 adoption coordinated through the Sustainable Banking and Finance Network
950 changes the relationship between firm-level access to finance and green invest-
951 ment behaviour, and whether any such moderation depends on the regulatory
952 capacity available to make the underlying policy credible. Using harmonized
953 World Bank Enterprise Survey data spanning 41 economies’ full 2018–2020
954 Green Economy Module rollout, a four-country waste-minimization supple-
955 mentary check, and an independently assembled 162-economy global sample
956 spanning 2021–2026 and reaching for the first time into high-income, non-

957 SBFN economies, we apply a deliberately escalating sequence of estimators—
958 a classical benchmark chosen to expose its own limitation, a Bayesian hier-
959 archical model built to give the moderation hypothesis a fair test, a causal
960 forest imposing no functional-form assumption at all, and, on the global
961 sample specifically, a country-fixed-effects specification the primary sample’s
962 single wave cannot support—and find four consistent results. First, access
963 to bank credit is robustly associated with a firm’s green practice adoption,
964 a relationship that survives every specification, subsample, and alternative-
965 outcome check in the paper and replicates independently in both the small
966 waste-minimization sample and the full 162-economy global sample. Second,
967 neither SBFN policy adoption nor regulatory quality moderates that rela-
968 tionship in any of the four estimators, on either sample, though both shift
969 the general level of green adoption directly and substantially in the primary
970 sample. Third, where genuine effect heterogeneity does exist, it is concen-
971 trated at the level of firm size rather than country institutions. Fourth, and
972 most importantly for the paper’s overall credibility, this entire pattern repli-
973 cates on a sample sharing almost no economies with the primary one, fielded
974 years later, measured with a different outcome item, and identified under a
975 materially sharper econometric strategy—the single strongest piece of evi-
976 dence in the paper that the null on institutional moderation is a real feature
977 of the data rather than an artefact of any one sample, period, or estimator.

978 For policy, the implication is not that green banking guidelines are inef-
979 fective, but that our evidence does not support treating them as an effective
980 lever specifically for widening the gap in green investment between financed
981 and unfunded firms—the margin most green-credit-policy literature implic-

982 itly assumes they operate on. If our tentative loan-tenor interpretation of
983 the overdraft-interaction finding is correct, policymakers designing or refining
984 SBFN-style frameworks may find more traction focusing supervisory atten-
985 tion on the term-lending instruments that actually finance green capital ex-
986 penditure, and on complementary measures—technical assistance, matching
987 grants, or credit guarantees—that address smaller firms’ absorptive-capacity
988 constraints directly, rather than assuming that a bank-level supervisory man-
989 date alone will reach firms regardless of scale.

990 We close with six limitations that bound how far these conclusions travel.
991 First, both samples are single cross-sections per country (Section 4 discusses
992 why a genuine panel on this outcome, and the staggered-adoption design it
993 would support, does not exist in the published WBES data): they cannot
994 rule out reverse causality (firms already predisposed to green investment may
995 find it easier to obtain credit) or unobserved, time-invariant country charac-
996 teristics correlated with both SBFN adoption and the baseline propensity to
997 adopt green practices, and our identification rests on firm-level variation in
998 credit access interacting with a plausibly firm-exogenous country-level policy
999 indicator, not a natural experiment; the global sample’s country fixed effects
1000 sharpen this only for the interaction terms, not for the credit main effect
1001 itself. Second, SBFN membership is a binary proxy for what the network’s
1002 own multi-tiered progression framework treats as a genuinely graded pro-
1003 cess; a natural extension would model policy stage—Commitment through
1004 Consolidating—as an ordinal or continuous treatment rather than collapsing
1005 it to membership status, which our binary coding necessarily does, and which
1006 for the global sample’s regional entries (the CEMAC and Eastern Caribbean

1007 groupings, Section 4) additionally rests on an inference about which specific
 1008 economies a regional listing covers rather than an explicit per-country state-
 1009 ment. Third, the four-economy waste-minimization check cannot re-test the
 1010 SBFN contrast itself, since three of its four economies are SBFN members,
 1011 and combines non-identical outcome items across countries by construction
 1012 of the WBES's own shortened, randomized module design in later survey
 1013 waves—a genuine data-availability constraint rather than a methodological
 1014 choice. Fourth, the global sample's CO2-monitoring outcome is a single item
 1015 rather than the primary sample's seven-item composite, and is a narrower
 1016 construct than "green practice adoption" in the sense the primary sample
 1017 measures it; that the same qualitative pattern holds on this narrower mea-
 1018 sure is reassuring but does not establish that it would hold on a richer one,
 1019 were a richer one available at global scale. Fifth, the Worldwide Gover-
 1020 nance Indicators' regulatory-quality measure is a broad, economy-wide gov-
 1021 ernance indicator rather than a banking-supervision-specific capacity mea-
 1022 sure; a supervisor-specific index, were one available consistently across this
 1023 country set, would sharpen the institutional-complementarity test consider-
 1024 ably. Sixth, our design tests only the intensive margin of the credit-to-green
 1025 channel (the adoption gap between financed and unfinanced firms); it cannot
 1026 rule out that SBFN policy operates instead on the extensive margin of which
 1027 firms obtain credit in the first place, nor can our firm-reported outcome
 1028 measure distinguish actual practice adoption from a compliance-reporting
 1029 gap between what firms report and what banks report to their supervisor
 1030 (Section 6). We leave each of these to future work.

1031 **Declaration of generative AI and AI-assisted technologies in the**
1032 **manuscript preparation process**

1033 During the preparation of this work the author(s) used Claude (An-
1034 thropic) to assist with data extraction and cleaning from World Bank Enter-
1035 prise Surveys microdata (including identification and extraction of the stan-
1036 dardized cross-country database underlying the global sample), construction
1037 of the analysis datasets and merge with country-level SBFN and Worldwide
1038 Governance Indicators data, execution of the statistical and machine-learning
1039 analyses reported in Section 5, generation of Figures 1–3, and drafting of por-
1040 tions of the manuscript text. After using this tool, the author(s) reviewed
1041 and edited the content as needed and take full responsibility for the content
1042 of the published article.

1043 **Appendix A. SBFN policy status and regulatory quality, full sam-**
1044 **ple**

1045 Table A.11 reports, for all 47 economy-years in the paper’s primary and
1046 small-sample supplementary check, the SBFN membership status coded for
1047 the analysis, the join year where applicable, the SBFN Multi-Tiered Progress
1048 Framework stage in force nearest the survey year, and the exact year of the
1049 Worldwide Governance Indicators observation used. Full per-country source
1050 citations (the specific SBFN Global Progress Report, country addendum, or
1051 accession announcement consulted for every determination) are available in
1052 the online supplementary material. The analogous 162-economy table for
1053 the global sample (Section 4) is summarized by region in Appendix C and

1054 provided in full in the online supplementary material and the replication
1055 package.

Table A.11: SBFN policy status and regulatory quality, full 47-
economy sample.

Economy	SBFN member	Join year	Stage at survey	WGI year
Albania 2019	0	—	None	2019
Armenia 2020	0	—	None	2020
Azerbaijan 2019	0	—	None	2019
Bangladesh 2022	1	2012	Advancing (ESG Integra- tion)	2022
Belarus 2018	0	—	None	2018
Bosnia and Herzegovina 2019	0	—	None	2019
Bulgaria 2019	0	—	None	2019
Croatia 2019	0	—	None	2019
Cyprus 2019	0	—	None	2019
Czechia 2019	0	—	None	2019
Egypt 2020	1	2016	Formulating (Prepara- tion)	2020
Estonia 2019	0	—	None	2019
Georgia 2019	1	2017	Developing (Implementa- tion)	2019
Greece 2018	0	—	None	2018
Hungary 2019	0	—	None	2019
India 2022	1	2016	Developing (Commit- ment)	2022

Economy	SBFN member	Join year	Stage at survey	WGI year
Indonesia 2023	1	2012	Consolidating (Maturing)	2023
Italy 2019	0	—	None	2019
Jordan 2019	1	2016	Commitment (Preparation)	2019
Kazakhstan 2019	0	—	None	2019
Kosovo 2019	0	—	None	2019
Kyrgyz Republic 2019	1	2018	Commitment (Preparation)	2019
Latvia 2019	0	—	None	2019
Lebanon 2019	0	—	None	2019
Lithuania 2019	0	—	None	2019
Malta 2019	0	—	None	2019
Moldova 2019	0	—	None	2019
Mongolia 2019	1	2012	Advancing (Implementation)	2019
Montenegro 2019	0	—	None	2019
Morocco 2019	1	2014	Advancing (Implementation)	2019
North Macedonia 2019	0	—	None	2019
Peru 2023	1	2013	Advancing (ESG Integration)	2023
Philippines 2023	1	2013	Advancing (ESG Integration)	2023
Poland 2019	0	—	None	2019
Portugal 2019	0	—	None	2019
Romania 2019	0	—	None	2019

Economy	SBFN member	Join year	Stage at survey	WGI year
Russia 2019	0	—	None	2019
Serbia 2019	0	—	None	2019
Slovak Republic 2019	0	—	None	2019
Slovenia 2019	0	—	None	2019
Tajikistan 2019	0	—	None	2019
Timor-Leste 2021	0	—	None	2021
Tunisia 2020	1	2019	Unverified	2020
Türkiye 2019	1	2015	Developing (Implementation)	2019
Ukraine 2019	0	—	None	2019
Uzbekistan 2019	0	—	None	2019
West Bank and Gaza 2019	0	—	None	2019

1056 Appendix B. Bayesian model convergence diagnostics

1057 Convergence diagnostics for every focal parameter of the hierarchical
1058 model reported in Table 4 are included directly in that table ($\hat{R}$ column); the
1059 full posterior summary also includes ess_{bulk} ranging from 797 (SBFN member)
1060 to 2,270 (credit $\times$ regulatory quality), comfortably above the conventional
1061 minimum threshold of 400 effective samples per parameter for stable poste-
1062 rior summaries at four chains. No divergent transitions were flagged during
1063 sampling (NUTS via the nutpie sampler, four chains of 1,000 post-warmup

1064 draws each after 1,000 tuning iterations).

1065 **Appendix C. Global-sample SBFN status, regional summary**

1066 Of the 162 global-sample economy-years, 61 are coded as SBFN members
1067 as of their survey year and 101 as non-members. Table C.12 reports the
1068 regional distribution of the 61 members, drawn from the SBFN data por-
1069 tal’s regional classification; five of the 61 (Peru, Philippines, Timor-Leste,
1070 Bangladesh, and Indonesia) carry the deeper per-country sourcing already
1071 reported in Appendix A, having been determined during the primary sam-
1072 ple’s original research pass rather than from the portal roster alone.

Table C.12: Global-sample SBFN members, by SBFN region.

Region	Member economy-years
Latin America and Caribbean	17
Africa	12
Europe and Central Asia	11
East Asia and Pacific	11
South Asia	6
Middle East and North Africa	4
Total	61

1073 **Additional material**

1074 *Online appendix*

1075 Appendix A, Appendix B, and Appendix C appear above; per-country
1076 source citations at the level of detail compiled during data construction (in-

dividual SBFN Global Progress Report page references, country-addendum
URLs, and IFC accession-announcement links for every non-obvious deter-
mination in the primary sample; the full 162-row global-sample SBFN/WGI
table underlying Appendix C) are available in the online supplementary
material accompanying this submission.

Data availability

Data in support of the findings of this study is available from the cor-
responding author upon reasonable request, subject to the World Bank En-
terprise Surveys' own terms of use for the underlying firm-level microdata
(enterprisesurveys.org, free registration required), including the standard-
ized cross-country database and Productivity Database underlying the global
sample. The country-level SBFN policy-status (including the SBFN data
portal's member roster underlying the global sample) and Worldwide Gover-
nance Indicators datasets constructed for this paper, together with the full
analysis and replication code, are available in a public GitHub repository;
the repository link is withheld from this anonymized manuscript, and repos-
itory access is restricted for the duration of double-anonymized review, to
avoid identifying the authors to reviewers. The link will be provided to the
editor upon submission and made publicly accessible again to readers upon
acceptance.

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